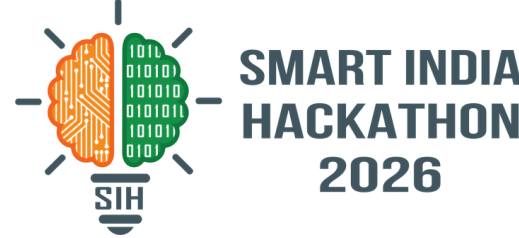




SMART INDIA HACKATHON 2026



Problem Statement ID– SIH26155

Problem Statement Title– AI-Driven Multi-Vendor Network Security Compliance Auditor

Organization name– National Technical Research Organisation (NTRO)

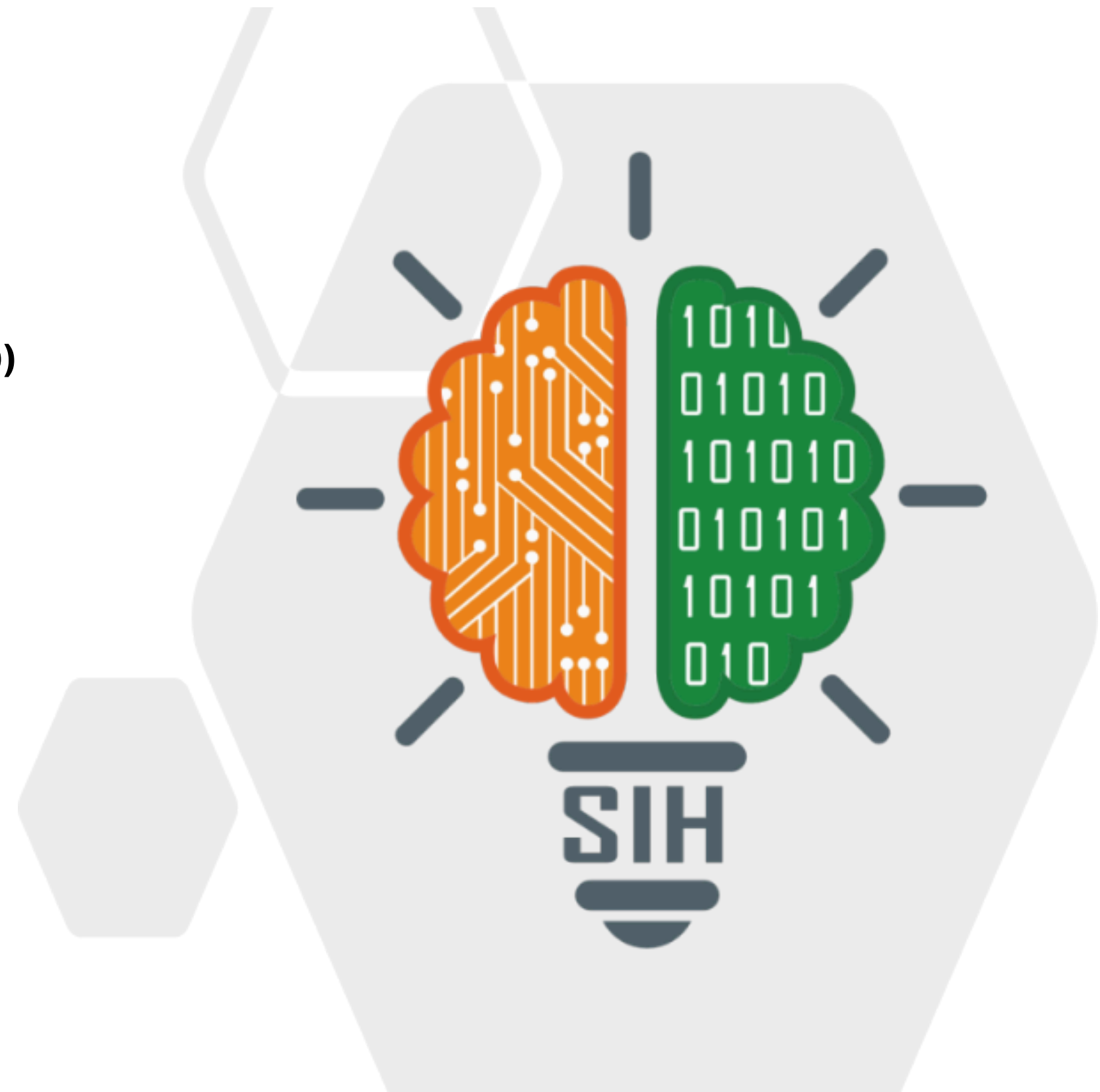
Theme– Blockchain & Cybersecurity

PS Category– Software

Team Name (Registered on portal)– Team Saarathi

Team ID–

Institution name– B. P. Poddar Institute of Management And Technology



Any vendor, uploaded or pulled live: 15 security checks, every verdict tied to the exact config line, new syntax learned from the admin with zero code changes.

What it does

- **Two ways in.** Upload CLI or JSON cloud rules, or pull running configs read-only over SSH.
- **Trusted detection.** Cisco IOS and FortiGate parsers used only if 70%+ of lines match.
- **Security Facts.** Each setting becomes value, scope, cited lines and assurance.
- **Actionable output.** Severity, CLI fix, one PDF per device with stated serial and OS.

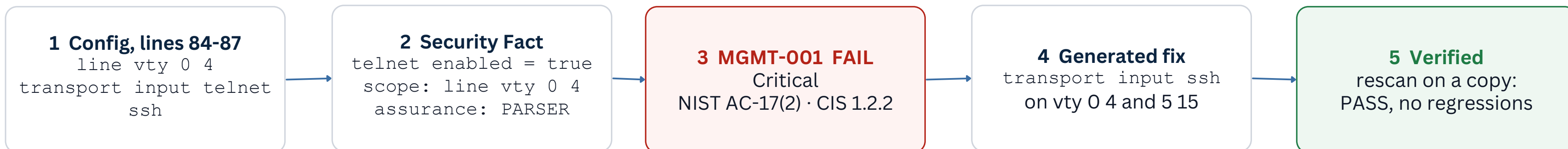
How it solves the PS

- **Syntax diversity:** rules read facts, never CLI text, so one rule serves every vendor.
- **Unseen vendors:** admin maps a raw line in Adaptive learning; a validated recognizer is saved.
- **Multi-framework:** NIST 800-53, DISA SRG, ISO 27001 or CIS: pick one at upload, or all four.
- **Safe remediation:** each fix is applied to a copy and rescanned before it is offered.

Why it is different

- **Honest coverage:** posture is always shown next to how much was actually decided.
- **AI proposes, humans decide:** AI answers stay provisional until confirmed.
- **Read-only by design:** collection only reads; no command is ever pushed to a device.
- **Day-one knowledge:** 97 shipped recognizers for 8 unparsed dialects + AWS security groups.

Worked example: one real line from our test file cisco_vulnerable.cfg, followed end to end

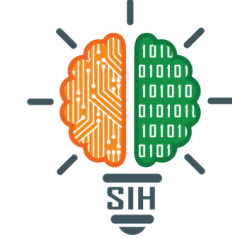




Saarthi

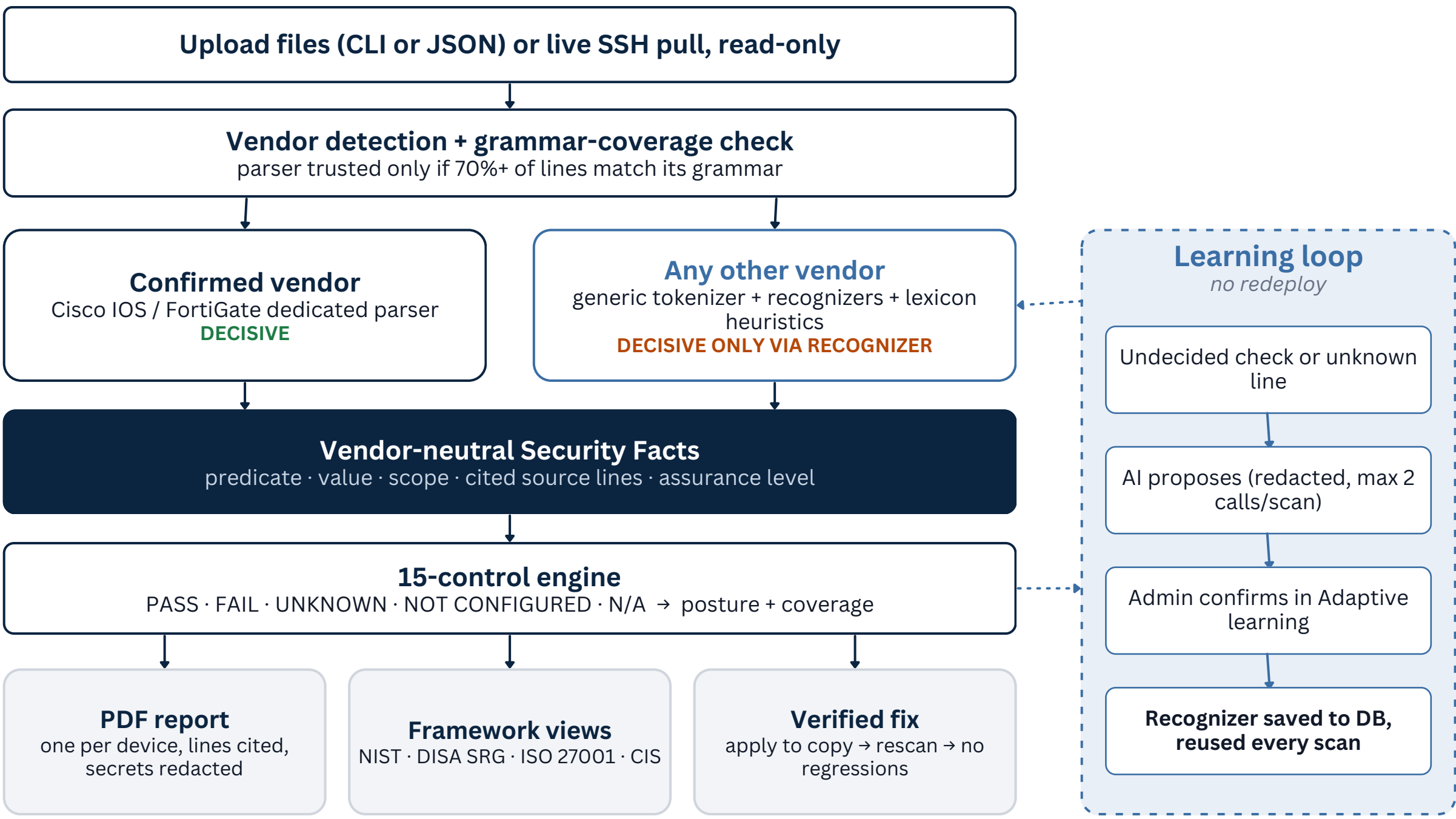


Technical Approach



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2026

Process flow



Decisive (scored): parser · confirmed · vendor default **Provisional (shown, never scored):** heuristic · AI-verified

Tech stack

Backend	Python · FastAPI · ReportLab PDF
Frontend	React 19 · Vite 6
Live collection	Netmiko / NAPALM, 12 platforms
Knowledge store	SQLite or PostgreSQL
AI (optional)	Groq · gpt-oss-120b, open-weight
Test suite	1,213 backend + 92 frontend passing

The 15 controls

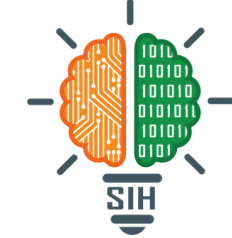
MGMT-001 Telnet off	MGMT-002 HTTP mgmt off	MGMT-003 Mgmt ACL
MGMT-004 SNMP community	MGMT-005 Password storage	MGMT-006 Idle timeout
MGMT-007 SSH v2 only	MGMT-008 AAA enabled	MGMT-009 Login banner
BOUNDARY-001 No any-to-any	BOUNDARY-002 Source routing off	BOUNDARY-003 CDP/LLDP on WAN
LOG-001 Remote syslog	LOG-002 Authenticated NTP	CRYPTO-001 Strong VPN crypto

■ Critical ■ High ■ Medium ■ Low

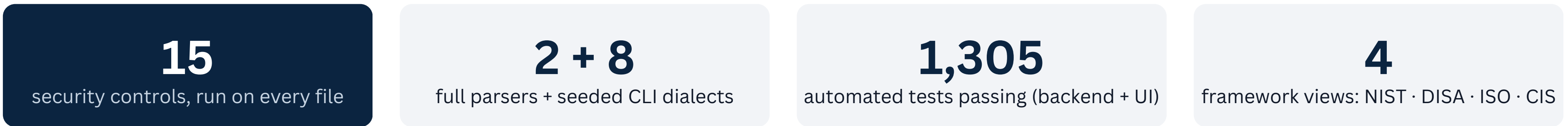
Why not just an LLM? One that guesses PASS on a line it never understood is worse than no auditor. AI is used only where rules cannot decide.



Feasibility and Viability



Feasibility: it already runs, end to end



Risks and how the design handles them

Firmware changes break parsers	Grammar-coverage gate: below 70% match the parser is not trusted and the file drops to the generic path. Unfamiliar firmware is flagged unverified.
AI hallucination and data leakage	AI is optional and sees only redacted, undecided checks. Answers must cite real lines and never move the score until a human confirms. Open-weight model, self-hostable.
Trust in automated fixes	Fix is applied to a copy, rescanned for target + no regression. Weak passwords, AAA lockout and any-to-any ACLs always go to a human.
Live SSH pull as an attack path	Read-only commands only. Private networks only, cloud metadata refused, the vetted IP is pinned against DNS tricks, and credentials are never stored or logged.



Viability

- **Near-zero cost:** open-source stack, public frameworks, runs on a laptop.
- **Buyers:** CII operators, PSUs, govt IT, MSPs and SOC's with mixed fleets.
- **Model:** on-prem licence for govt, SaaS for MSPs; knowledge grows per deployment.



Roadmap

- | | |
|-------|---|
| NEXT | In-app assistant chat (API already live); UI sends the API key; users and roles |
| THEN | Self-hosted LLM; admin-accounting and management-TLS controls |
| SCALE | Promote PAN-OS and Junos to full parsers; Azure NSG and SONiC knowledge |



Impact and Benefits

Measured on cisco_vulnerable.cfg

0 → 72

posture after one verified fix pass, given 3 admin inputs

Before  0

After  72

Left for a human on purpose: MGMT-005 weak passwords, MGMT-008 AAA lockout risk, BOUNDARY-001 any-to-any ACL.

Honest by default

FortiGate: password storage is reported **critical, not assessed** instead of a fake PASS.

Unknown vendor: posture shows "-" and coverage 0% until something is actually decided.

Never taught, still useful: Junos 28%, Huawei 31%, PAN-OS 34% coverage out of the box.

Who benefits



Network admins

One upload or live pull replaces a per-vendor checklist. Fixes arrive verified on a copy.



Security / SOC teams

Telnet, default SNMP, open HTTP management, any-to-any rules and AWS groups open to 0.0.0.0/0.



Compliance auditors

PDF per device, assessed against the framework chosen at upload. MGMT-001 maps to NIST AC-17(2), CIS 1.2.2.



Govt & CII (NTRO, NCIIPC)

Upload mode runs fully offline; live pull stays inside private networks. No vendor lock-in.



Security

Closes the misconfigurations attackers try first, before they are exploited, across every vendor in the fleet.



Economic

Free open stack instead of per-device commercial suites; less spend on external audits.



Strategic

Indigenous, vendor-neutral and auditable logic that can run inside air-gapped networks.

References

Frameworks (exact versions mapped)

- CIS Cisco IOS XE 17.x Benchmark v2.2.1
- CIS FortiGate 7.4.x Benchmark v1.0.1
- NIST SP 800-53 Rev. 5 (OSCAL catalog 5.2.0)
- DISA Network Device Management SRG V4
- ISO/IEC 27001:2022 Annex A

Vendor documentation

- Cisco IOS / IOS-XE command references
- Fortinet FortiOS CLI reference
- Guides for the 8 seeded dialects: EOS, Junos, VRP, PAN-OS, Gaia, EXOS, RouterOS, AOS-CX

Built with

- FastAPI, Pydantic, React 19, Vite, ReportLab
- Netmiko, NAPALM, SQLite / PostgreSQL
- Groq API (gpt-oss-120b)

Source code

- github.com/astrilo-monk/sih-26155

Deliverables:

- <https://26155-deliverables.vercel.app/>

SCAN ME !



<https://netaudit-three.vercel.app/>

Scan To View
MVP

Research workflow

1

Study the frameworks

CIS, NIST 800-53, DISA SRG and ISO 27001 control wording

2

Collect real configs

10 CLI dialects, cloud JSON, and look-alikes that fool naive detectors

3

Design the fact schema

vendor-neutral predicates with scope, evidence and assurance

4

Build and test

15 controls backed by 1,305 automated tests

5

Validate

paired vulnerable vs secure configs; every fix rescanned